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Journal Name

COMMUNICATION

Chemically exfoliated boron nitride nanosheets induced robust interfacial layers for stable solid state Li metal batteries

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The two dimensional (2D) boron nitride nanosheets (BNNs) were chemically exfoliated from bulk boron nitride and coated onto the surface of poly(ethylene oxide) (PEO)-based electrolyte through a dry-pressing transfer process. The fabricated BNNs coating can form a robust interfacial layer to improve the chemical and mechanical stability of the PEO-based electrolyte leading to enhanced performance of solid state Li metal batteries.

Rechargeable Li metal batteries with high energy densities, long span-life, and excellent safety are highly desirable for future electrical vehicles and grid-scale stationary energy storages.^{1, 2} However, utilization of Li metal anode in liquid electrolyte is long term hindered by the low Coulombic efficiency and the growth of Li dendrites during electrochemical plating/stripping processes.^{3, 4} Comparing to these limitations of conventional liquid electrolytes along with potential safety risks,^{5, 6} the advantages of solid state electrolytes are remarkable, such as high shear modulus,⁷ high Li transference number,⁸ and non-inflammable.⁹

Among the developed solid state electrolytes, the solid polymer electrolytes (SPEs) are attractive because of their high flexibility, good processability, as well as low interface resistance with Li metal anode.^{10, 11} Especially, PEO based electrolyte has caught the most attention since it was first reported by Wright in 1970s.^{12, 13} However, the PEO based electrolyte shows a poor ionic conductivity at room temperature, which has to be operated at an elevated temperature and would lose the mechanical strength to suppress the growth of Li dendrites.^{10, 14, 15} Regarding to this, studies about the improvement of mechanical strength and ionic conductivity of PEO based composite SPEs at room temperature have been comprehensively conducted in the past decades.^{13, 16, 17} For examples, cross-linked polymers SPEs¹⁸⁻²⁰,

nanoparticle fillers composite SPEs^{13, 21} and three-dimensional framework structures reinforced SPEs^{16, 22, 23} have been developed to suppress the growth of Li dendrites in PEO electrolytes based Li metal batteries. In contrast to the rapid development of mechanical strength reinforcement of PEO electrolytes, the interfacial design to suppress Li dendrites growth is rarely considered for the PEO electrolytes.

Herein, we proposed the chemically exfoliated BNNs protection layer on the surface of PEO based electrolyte to suppress the Li dendrites growth in solid state Li metal batteries. As shown in Figure 1a, the BNNs were synthesized through two steps. In the first step, the lithium intercalated h-BN was obtained by inserting lithium atoms into the layer of h-BN. In the second step, the lithium intercalated h-BN react violently with H₂O to exfoliate and disintegrate the h-BN.²⁴ This method can be also used to prepare MoS₂ nanosheets (experimental section and Fig. S1a-d, ESI†). The typical transmission electron microscopy (TEM) image of as-synthesized BNNs (Fig. 1b) shows the ultrathin 2D and flexible sheet morphologies, indicating the successful exfoliation of BN. The high resolution TEM (HRTEM) image (Fig. 1c) further reveals the lattice spacing of BN nanosheet as 0.25 nm.^{25, 26} Furthermore, the influence of reaction time of Li infusion into BN flakes on the morphologies of unexfoliated flakes was investigated by scanning electron microscopy (SEM) to show the breakage of h-BN flakes during the Li infusion process. As shown in Fig. S2a-d (ESI†), the sizes of unexfoliated flakes gradually decreased from 10 μm to 1 μm with prolonging the Li infusion reaction time from 0 hours to 12 hours, indicating that the Li is highly reactive to BN leading to the following exfoliation of BNNs. TEM images of as-obtained BNNs at different Li infusion reaction time are also shown in Fig. S2e-h (ESI†), which indicates that the size of BNNs also decreased with prolonging the Li infusion time. The thickness of BNNs was further measured by atomic force microscopy (AFM) as shown in Fig. S3a (ESI†). The height variation line in Fig. S3b (ESI†) suggests that the thickness of BNNs is around 2-3 nm, indicating the atomic thickness feature of as-obtained BNNs. The BNNs obtained at the Li infusion time of 2 hours were used for the following characterizations and studies due to their suitable sizes and thicknesses.

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* Footnotes relating to the title and/or authors should appear here.

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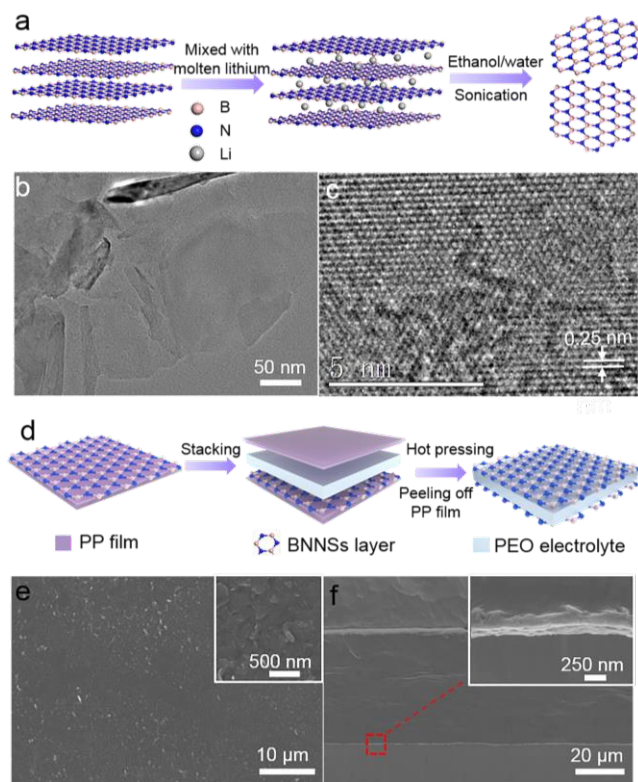


Fig. 1 (a) Schematic illustration of the BNNs exfoliation process. (b) TEM image of BNNs. (c) HRTEM image of BNNs. (d) Schematic illustration of the transfer process. (e, f) SEM image for the surface and cross-section of BNNs-coated PEO electrolyte (inset: corresponding magnified SEM image).

As shown in Fig. 1d, the obtained BNNs were assembled onto a polypropylene (PP) separator via a vacuum assisted filtration process to form a BNNs coated PP separator. Then the BNNs coated PP separators are stacked with PEO electrolyte and hot-pressed together to transfer BNNs onto the surface of PEO electrolyte. After that, the PP separators are peeled off and the BNNs double-layer coated PEO electrolyte is formed. The photos in Fig. S4 (ESI†) show that the PEO-based electrolyte became a little opaque after the BNNs coating transfer. The surface SEM image of BNNs coated PEO electrolyte (Fig. 1e) shows that the stacked nanosheets structure appeared in contrast to the smooth surface of pristine PEO electrolyte (Fig. S5a, ESI†). In addition, the stacked BNNs were formed on the surface of PEO electrolyte regardless of the mass of BNNs that transferred onto the surface of PEO electrolyte (Fig. S5b, c, e, f and g, ESI†), indicating the good efficiency of as-developed BNNs assembly-hot press transfer route. Moreover, the cross-sectional SEM images of BNNs-coated PEO electrolyte (Fig. 1f) shows that a continuous dense layer formed on the surface of PEO electrolyte with a thickness of 150 nm when the mass loading of BNNs was 0.02 mg/cm² (measured by mass change of the BNNs coated PP separators before and after transferring BNNs onto the PEO electrolyte).^{27, 28} This dense BNNs coating layer on the surface of PEO electrolyte will endow it good mechanical stability and homogeneous Li⁺ ion flux distribution.

The ionic conductivity of BNNs-coated PEO electrolyte with different mass loading of BNNs were measured according to

electrochemical impedance spectroscopy (EIS) method at first. The ionic conductivity of BNNs-coated PEO electrolyte was calculated based on the resistance obtained by the results of EIS. Fig. 2a presents the temperature dependence of ionic conductivities for the different mass loadings of BNNs-coated PEO electrolytes at a temperature range from 30 °C to 100 °C. The ionic conductivities of BNNs-coated PEO electrolytes with the mass loading of 0.02 and 0.036 mg/cm² are similar to that of pristine PEO electrolyte, the ionic conductivity of which can reach up to 2.0×10^{-4} S/cm at 60 °C. Based on the good ionic conductivity of as-fabricated BNNs-coated PEO electrolyte, the cycling stability of symmetric Li/BNNs-coated PEO/Li cell was further evaluated. The symmetric cells were cycled at 60 °C by one hour charging, ten minutes resting and one hour discharging at a current density of 0.3 mA/cm². As shown in Fig. 2b and c, the time-dependent voltage profiles for the cells with 0.02, 0.036 and 0.05 mg/cm² of BNNs-coated electrolytes retained stable over 350 h cycling, whereas the cell based on pristine PEO electrolyte experienced a sharp voltage drop after only 140 h cycling indicating a short circuit of the Li/PEO/Li cell. More impressively, the Li/Li symmetric cell with 0.11 mg/cm² of BNNs-coated electrolyte can be stably cycled for 750 hours (Fig. 2c), which is more than 5-fold of the cycling life of the cell with pristine PEO electrolyte. In addition, the Li plating/stripping polarization voltage (the difference between Li plating voltage and stripping voltage) gradually increased from 180 mV to 240 mV with the increase of mass loading of BNNs from 0.02 mg/cm² to 0.11 mg/cm² (insets in Fig. 2b and c). This result is consistent with the variation trend of ionic conductivity of BNNs-coated PEO electrolyte with the change of mass loading of BNNs (Fig. 2a). The BNNs prepared by urea-assisted ball milling method²⁹ was also used to fabricate the BNNs-coated PEO electrolyte, which showed slightly worse cycling performance than the chemically exfoliated BNNs-coated PEO electrolyte (Fig. S6, ESI†). To minimize the polarization effect of BNNs layer coating on Li plating/stripping, the following electrochemical performance evaluation of electrolytes and batteries were all based on the

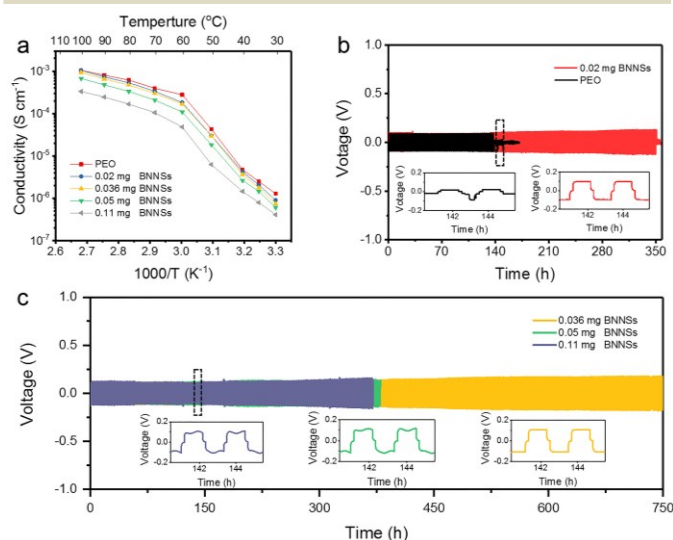


Fig. 2 (a) The ionic conductivity of BNNs-coated PEO electrolyte. (b, c) Galvanostatic cycling curves of Li/Li symmetrical cells based on PEO and the different mass loading BNNs-coated PEO electrolyte. The cells were test at a current density of 0.3 mA cm⁻² and 60 °C.

0.02 mg/cm² loading of BNNSs-coated PEO electrolyte and controlled pristine PEO electrolyte.

To show the interfacial resistance between the electrolyte and Li metal anode, the EISs were measured on the Li/PEO/Li cell and Li/BNNSs-coated PEO/Li cell before and after 100 h cycling. As shown in Fig. 7a and b (ESI[†]), the interfacial resistance of Li/PEO/Li cell increased 46 Ω much larger than the increase of interfacial resistance of Li/BNNSs-coated PEO/Li cell (20 Ω), indicating that the interface between PEO and Li deteriorated much faster than the interface of BNNSs-coated PEO/Li. To further verify the stability of electrolyte/Li interfaces, the surface and cross-sectional morphologies of the Li metal anode and Li/electrolyte/Li cell were characterized by SEM after 200 h cycling. As shown in Fig. 3a, the surface of Li metal anode cycled with the PEO electrolyte exhibited extensively mossy Li dendrites. In contrast, the Li metal anode cycled with the BNNSs-coated electrolyte displayed a much more smooth surface (Fig. 3d), which means that the Li dendrites were heavily suppressed by the BNNSs-coated electrolyte. The cross-sectional SEM characterizations of sandwiched Li/electrolyte/Li (Fig. 3b, c, e, and f) further revealed the interfacial variation of the cell after 200 h cycling. Fig. 3b shows that Li dendrites grow along the interface of Li metal anode and PEO electrolyte. The corresponding magnified SEM image (Fig. 3c) clearly shows that the PEO electrolyte was pierced by Li dendrites leading to the short circuit of Li/PEO/Li cell. Whereas, the boundary between the BNNSs-coated PEO electrolyte and Li metal anode is much more uniform than that of PEO/Li (Fig. 3e). The magnified SEM image (Fig. 3f) further reveals tightly contacted interface of BNNSs-coated PEO and Li metal anode without any Li protruding. These morphological characterizations provide good evidences that the growth of Li dendrites is successfully suppressed by the BNNSs-coated PEO electrolyte during the Li plating/stripping cycles. The mechanical strength of BNNSs-coated PEO electrolyte is better than PEO electrolyte at 60 $^{\circ}\text{C}$ (Fig.S8, ESI[†]). The different mechanisms of electrochemical Li plating/stripping with PEO electrolyte and BNNSs-coated PEO electrolyte are schematically illustrated in Fig.S9 (ESI[†]). In a typical Li/PEO/Li symmetric cell, Li dendrites can be promoted and eventually penetrate the PEO electrolyte due to its low mechanical strength at 60 $^{\circ}\text{C}$ and the inhomogeneous distribution of Li⁺ ion flux. In contrast, the multilayered BNNSs coating layer can result in a uniform plating/stripping of Li due to its improved mechanical strength and a more homogeneous distribution of Li⁺ ion flux.

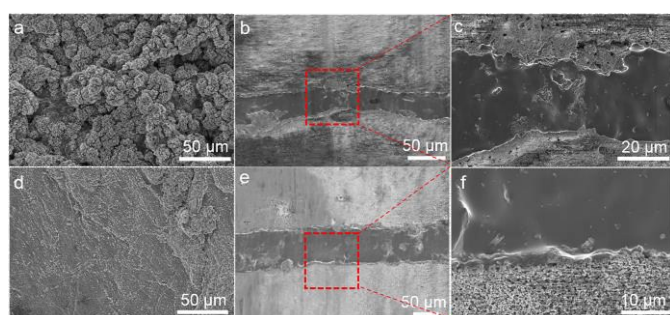


Fig. 3 (a) The surface SEM image of Li electrode of the Li/PEO/Li cell. (b, c) The cross-sectional SEM images of the Li/PEO/Li cell. (d) The surface SEM image of Li electrode of the Li/BNNSs-coated PEO/Li cell. (e, f) The cross-sectional SEM image of the Li/BNNSs-coated PEO/Li cell.

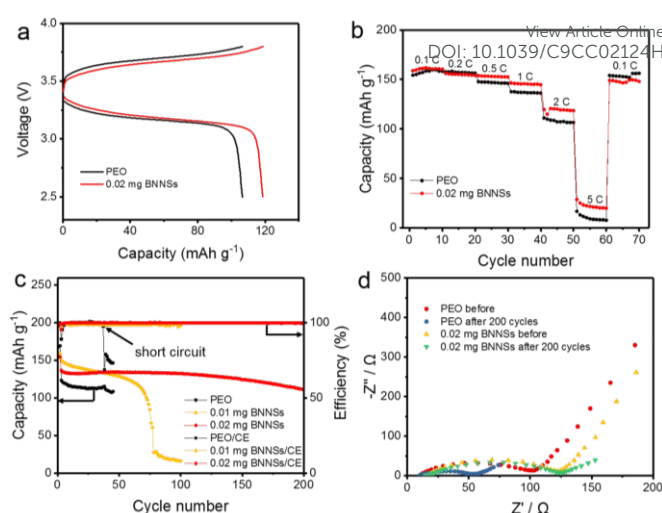


Fig. 4 (a) Typical charge and discharge voltage profiles of Li/LFP cells with PEO and BNNSs-coated PEO electrolyte at 2 C. (b) The Rate performance comparison of the Li/PEO/LFP cell and Li/BNNSs-coated PEO/LFP cell. (c) The plots of specific discharge capacity and Coulombic efficiency versus cycle number of the Li/PEO/LFP cell and Li/BNNSs-coated PEO/LFP cell cycling at 2 C. (d) EISs of the Li/PEO/LFP and Li/BNNSs-coated PEO/LFP cells before and after 200 cycles at 2 C. The cells were tested at 60 $^{\circ}\text{C}$.

All solid-state Li metal batteries were assembled to evaluate their electrochemical performances. Fig. 4a shows typical charge-discharge voltage profiles of Li/LFP cells at 2 C with the PEO electrolyte and BNNSs-coated PEO electrolyte. These profiles exhibited the normal voltage plateaus of LFP cells demonstrating that the BNNSs-coated PEO electrolyte can be used in solid-state Li metal batteries.³⁰ The rate performance of Li/PEO/Li and Li/BNNSs-coated PEO/LFP cells were also tested and compared. The charge/discharge voltage profiles at different rates (Fig. S10, ESI[†]) indicates that the charge-discharge overpotentials of the cell increased with the increase of current density caused by the limited diffusion rate of Li⁺ ion in the solid PEO electrolyte.⁹ The plot of discharge capacity versus cycling number (Fig. 4b) shows that the cell with BNNSs-coated PEO electrolyte delivered the specific discharge capacities of 160, 154, 152, 145, 119 and 20 mAh g⁻¹ at 0.1 C, 0.2 C, 0.5 C, 1 C, 2 C, and 5 C, respectively, whereas the values of the cell with PEO electrolyte are only 158, 156, 146, 136, 106 and 8 mAh g⁻¹, respectively. The results demonstrated that the cell with BNNSs-coated PEO electrolyte displayed much better rate performance than that with PEO electrolyte.

As to the long-term cycling performance, the solid Li/LFP cells were pre-activated at 0.1 C for two cycles before the cycling test. As shown in Fig. 4c, 0.02 mg/cm² the BNNSs-coated PEO electrolyte based cell delivered the specific discharge capacity of 110 mAh g⁻¹ over 200 cycles with a very slow decay rate (0.09% per cycle) at the rate of 2 C, whereas the PEO electrolyte based cell short-circuited at the 40th cycle. Due to the incompletely coated on the surface of PEO electrolyte (Fig.S5c, ESI[†]), the 0.01 mg/cm² BNNSs-coated PEO electrolyte based cell experienced a capacity drop at the 75th cycle. The average Coulombic efficiency is 99.9 $\pm$ 0.1% for a Li/LFP cell using 0.02 mg/cm² BNNSs-coated PEO electrolyte during the 200 cycles. The improved performance of Li/LFP cell with BNNSs-coated PEO electrolyte can be attributed to the dendrites growth suppression and stable

interface of Li/BNNSSs-coated PEO electrolyte. The EIS was further used to evaluate the interfacial impedance of the Li/electrolyte/LFP cells before and after 200 cycles. As shown in Fig. 4d, before cycling, the interfacial impedance of the cell with the BNNSSs-coated PEO electrolyte (108 Ω) is a little higher than that of the cell with the PEO electrolyte (98 Ω) due to the polarization induced by BNNSSs coating. After 200 cycles, the interfacial impedance for the cell with PEO electrolyte decreased to a small value (48 Ω) due to the short circuit of the cell. However, it is apparent that the cell with BNNSSs-coated PEO electrolyte displayed an interfacial impedance with negligible variation after 200 cycles. This phenomenon suggests that the interface between the electrode and BNNSSs-coated PEO electrolyte remains relatively stable and robust contact, resulting in the excellent cycling stability of Li/BNNSSs-coated PEO/LFP cell.

In summary, we demonstrated that an ultrathin chemically exfoliated BNNSSs layer coating can form robust protection for the PEO electrolyte to improve the performance of solid state Li metal batteries. Our study showed that the BNNSSs-coated PEO electrolyte exhibited superior resistance to the growth of Li dendrites growth in comparison to pristine PEO electrolyte. As a result, all solid state Li/LFP cell with the BNNSSs-coated PEO electrolyte delivered the specific discharge capacity of 110 mAh g⁻¹ at 2 C over 200 cycles. The proposed atomic 2D material coating strategy for the protection of solid polymer electrolyte will open a new avenue to improve the performance of solid state Li metal batteries.

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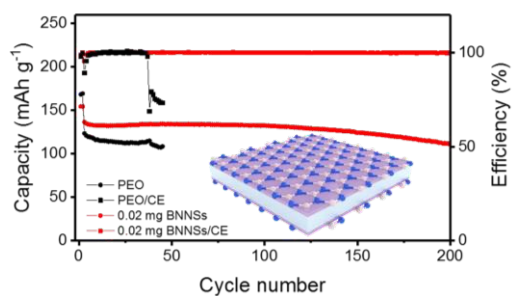
Conflicts of interest

There are no conflicts to declare.

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Graphic Abstract:



The BNNSs coating layer can form robust protection for the PEO-based electrolyte to suppress the growth of Li dendrites and improve the performance of solid state Li metal batteries.